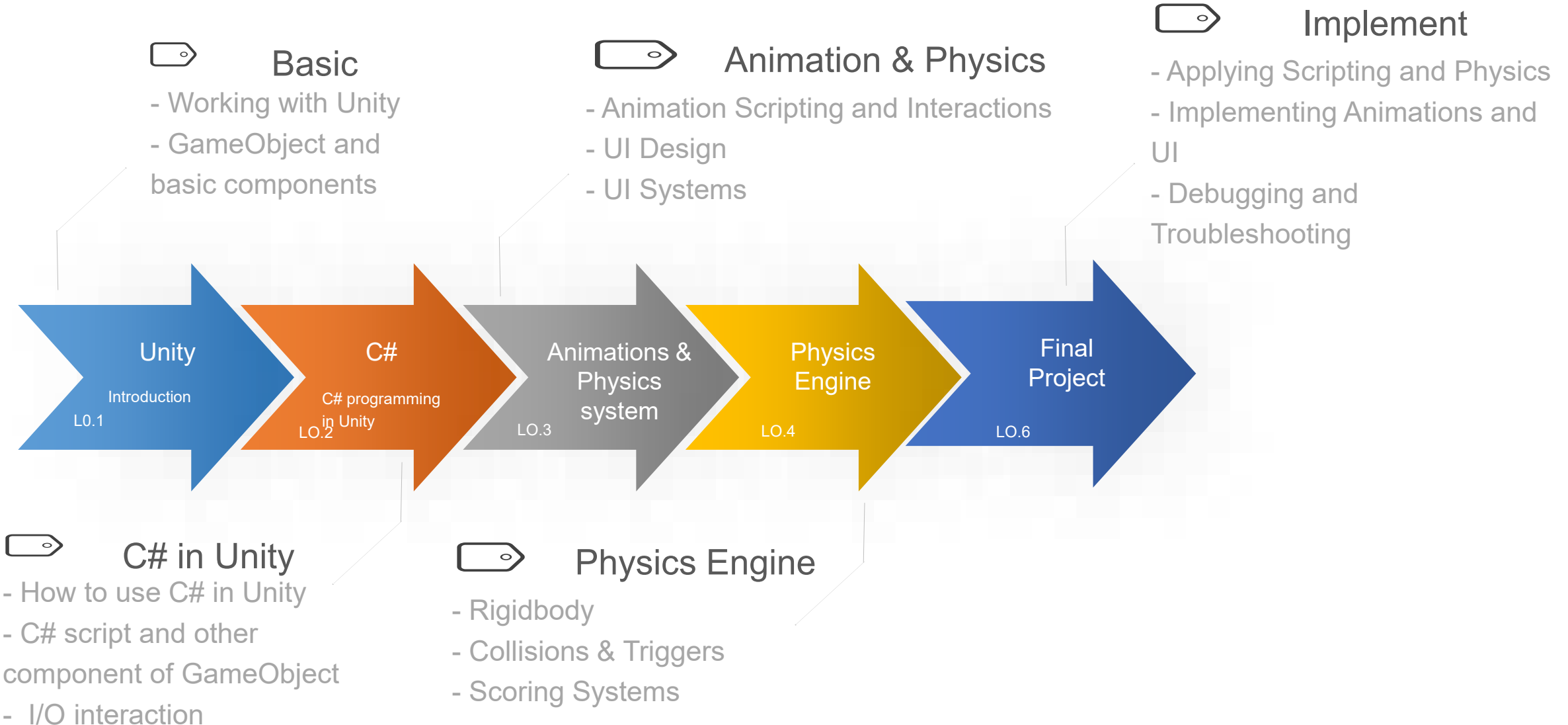


Introduction to Unity

**GAME PROGRAMMING
C# PROGRAMMING AND UNITY**

What you will learn in this Course



Game

- Game is a structured form of play, usually undertaken for entertainment or fun, and sometimes used as an educational tool. Many games are also considered to be work or art
- Games can take many forms, including physical activities, board games, card games, video games, sports...
- They can be played alone, in teams, offline or online; by amateurs or by professionals
- Key components of games are goals, rules, challenge, and interaction

Video game

- Video games are interactive electronic games played on various devices, including computers, gaming consoles, smartphones, tablets, and handheld gaming devices
- They are a form of digital entertainment that allows players to engage with virtual worlds, characters, and challenges through input devices like keyboards, mice, controllers, or touchscreens



Game engine

- A software framework or platform designed to create and develop video games
- It provides developers with a set of tools, libraries, and functionalities that simplify and expedite the game development process
- Some popular game engines:
 - Unity
 - Unreal Engine
 - Godot Engine
 - CryEngine
 - Lumberyard
 - ...



Game engine

- The most popular game engines contain some or all of the following functionality:
 - Graphics rendering engine, supporting 2D or 3D graphics
 - Physics engine that supports collision detection
 - Audio engine to load and play sounds and music files
 - Scripting support to implement gameplay logic
 - World object model defining the contents and properties of the game world
 - Animation handling to load animation frames and play them

Game engine

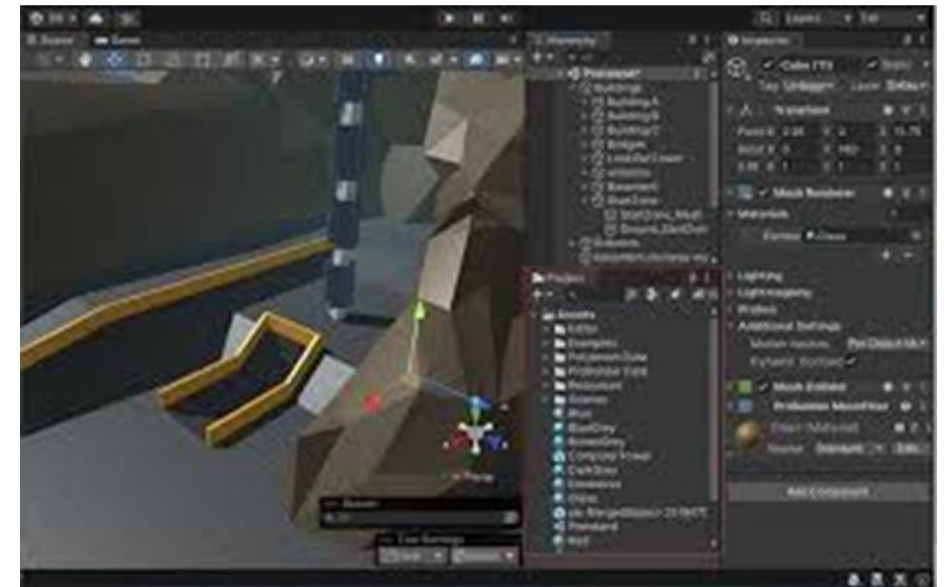
- The most popular game engines contain some or all of the following functionality:
 - Networking code to allow for multiplayer, downloadable content, and leaderboards
 - Multithreading to allow game logic to execute simultaneously
 - Memory management because no computer has unlimited memory
 - Artificial intelligence for pathfinding and computer opponents

Introduction to Unity

- Unity is a cross-platform game engine initially released by Unity Technologies, in 2005. The focus of Unity lies in the development of both 2D and 3D games and interactive content
- Unity now supports over 20 different target platforms for deploying, while its most popular platforms are the PC, Android and iOS systems
- Unity features a complete toolkit for designing and building games, including interfaces for graphics, audio, and level-building tools, requiring minimal use of external programs to work on projects

Introduction to Unity

- **Using of Unity**
 - Game development
 - Interactive Experiences
 - Previsualization for Film
 - Architectural Visualization
 - Animation
 - Simulations



Cross-Platform Capabilities

Unity's cross-platform capabilities empower developers to create games and applications that can seamlessly run on multiple platforms, including mobile devices (iOS, Android), consoles, desktops (Windows, macOS), web browsers, AR/VR devices, and more

Unparalleled platform support

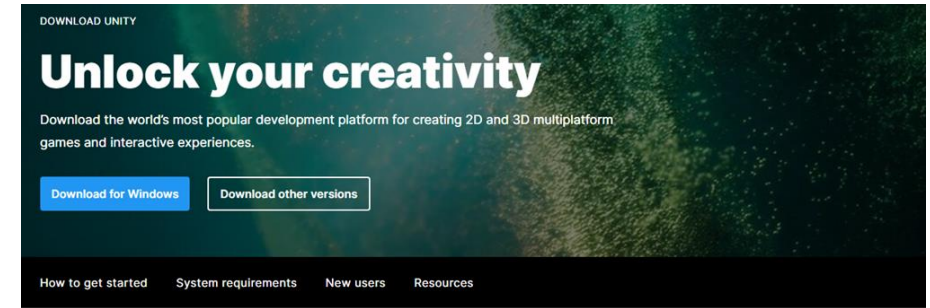


Installation

- Download Unity Hub:

<https://unity.com/download>

- Install Unity Hub
- Run Unity Hub
- Create account / Sign in
- Install / Choose Unity Editor:
Visual Studio 2019 or later



Create with Unity in three steps

1. Download the Unity Hub

Follow the instructions onscreen for guidance through the installation process and setup.

[Download for Windows](#)
[Download for Mac](#)
[Download for Linux \(beta\)](#)

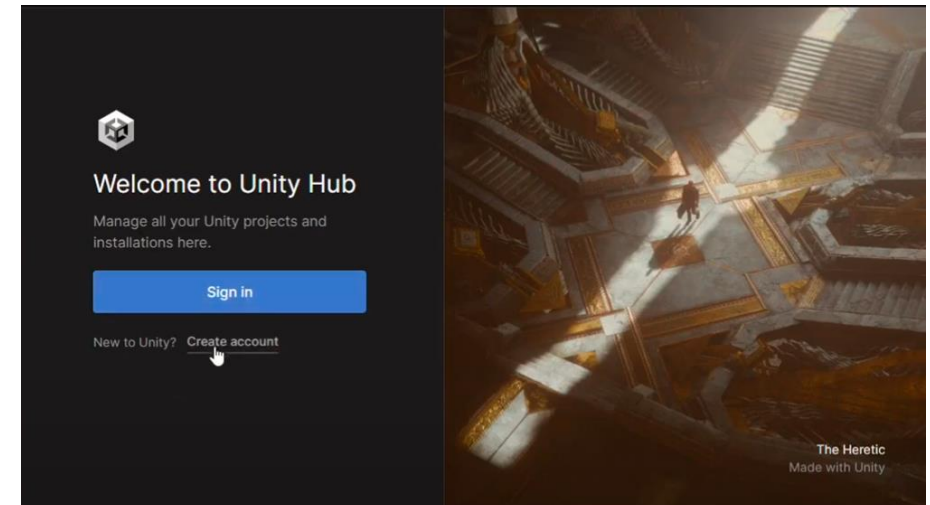
2. Choose your Unity version

Install the latest version of Unity, an older release, or a beta featuring the latest in-development features.

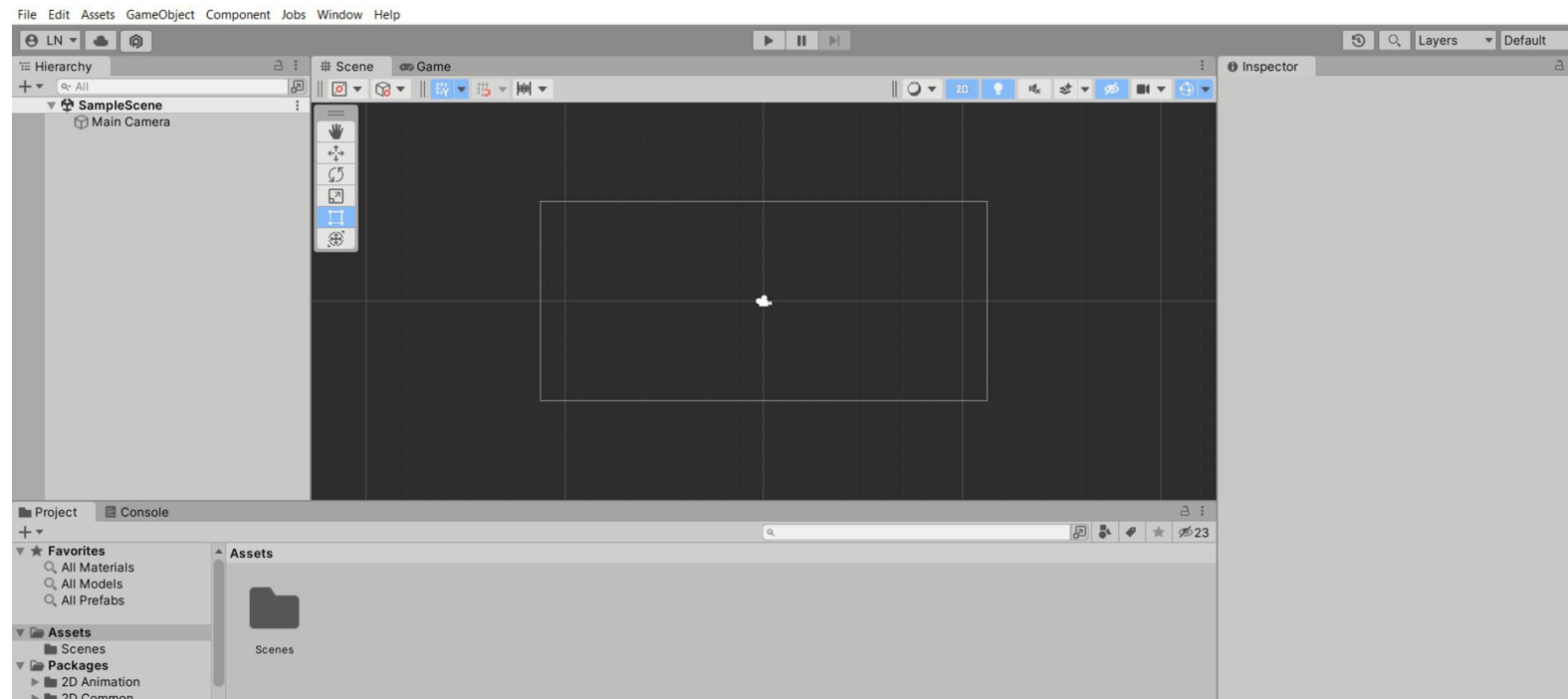
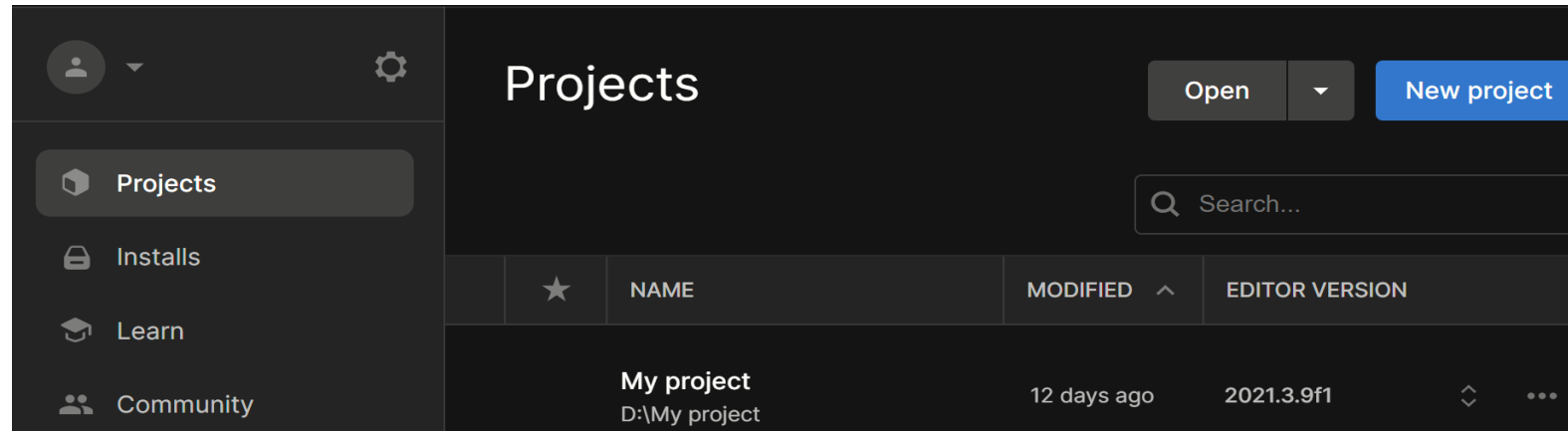
[Visit the download archive](#)

3. Start your project

Begin creating from scratch, or pick a template to get your first project up and running quickly. Access tutorial videos designed to support creators, from beginners to experts.



Installation

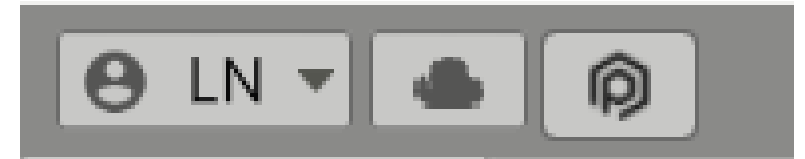


Unity IDE Interface



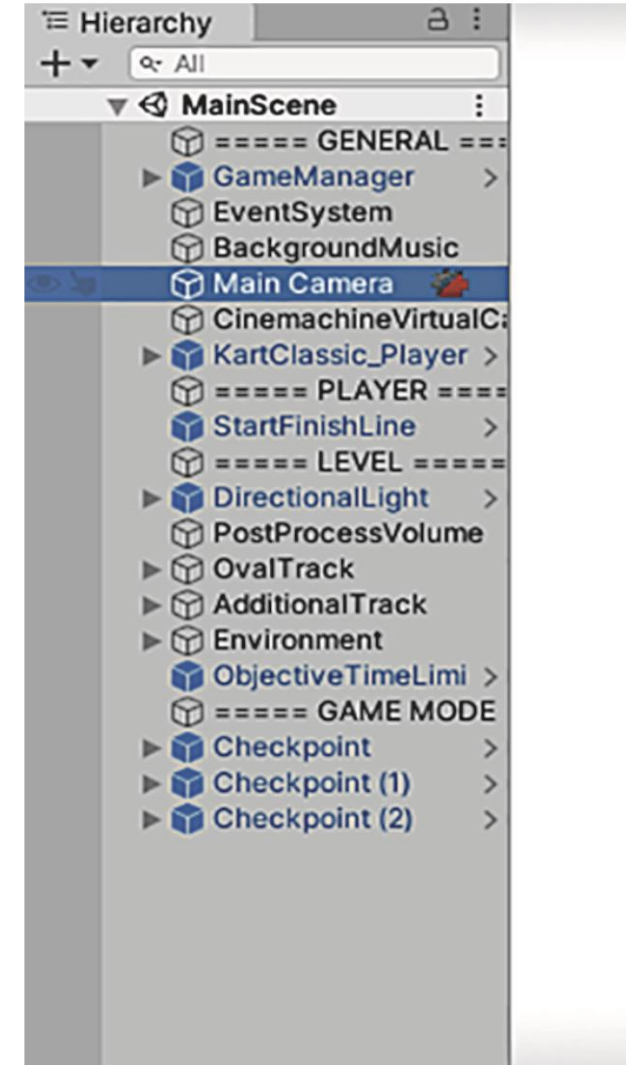
Unity IDE Interface

- **(A) The Toolbar** provides access to your Unity Account and Unity Cloud Services. It also contains controls for Play mode; Undo history; Unity Search; a layer visibility menu; and the Editor layout menu



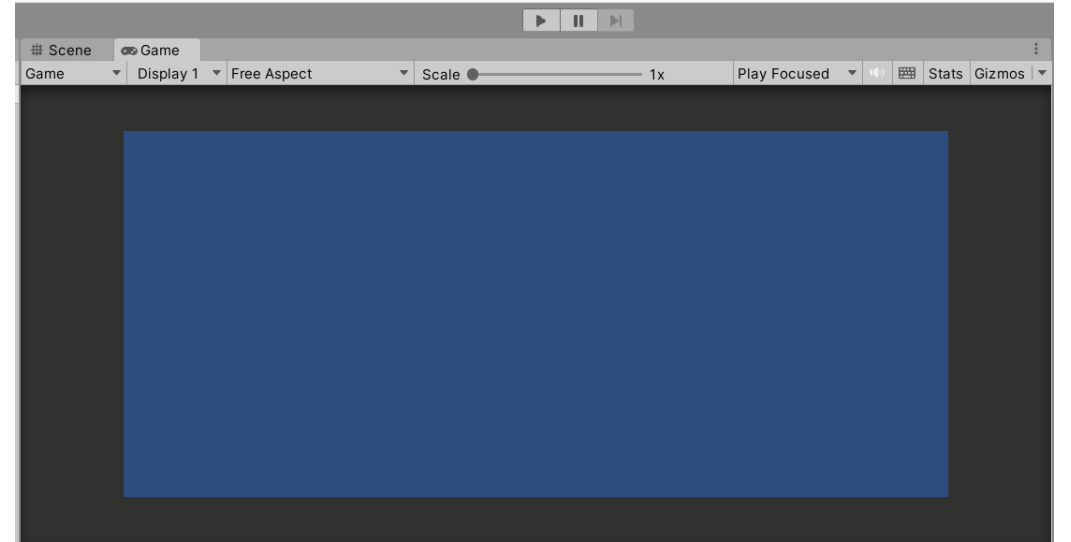
Unity IDE Interface

- **(B) The Hierarchy window** is a hierarchical text representation of every **GameObject** in the **Scene**
 - **Hierarchy** lists the name of every object in the scene and nests the names together according to their hierarchy linkages in the scene
 - Basically, it's a way of selecting objects by name instead of hunting them down and clicking them within the Scene view
 - The Hierarchy linkages group objects together visually, like folders, allowing you to move the entire group as one



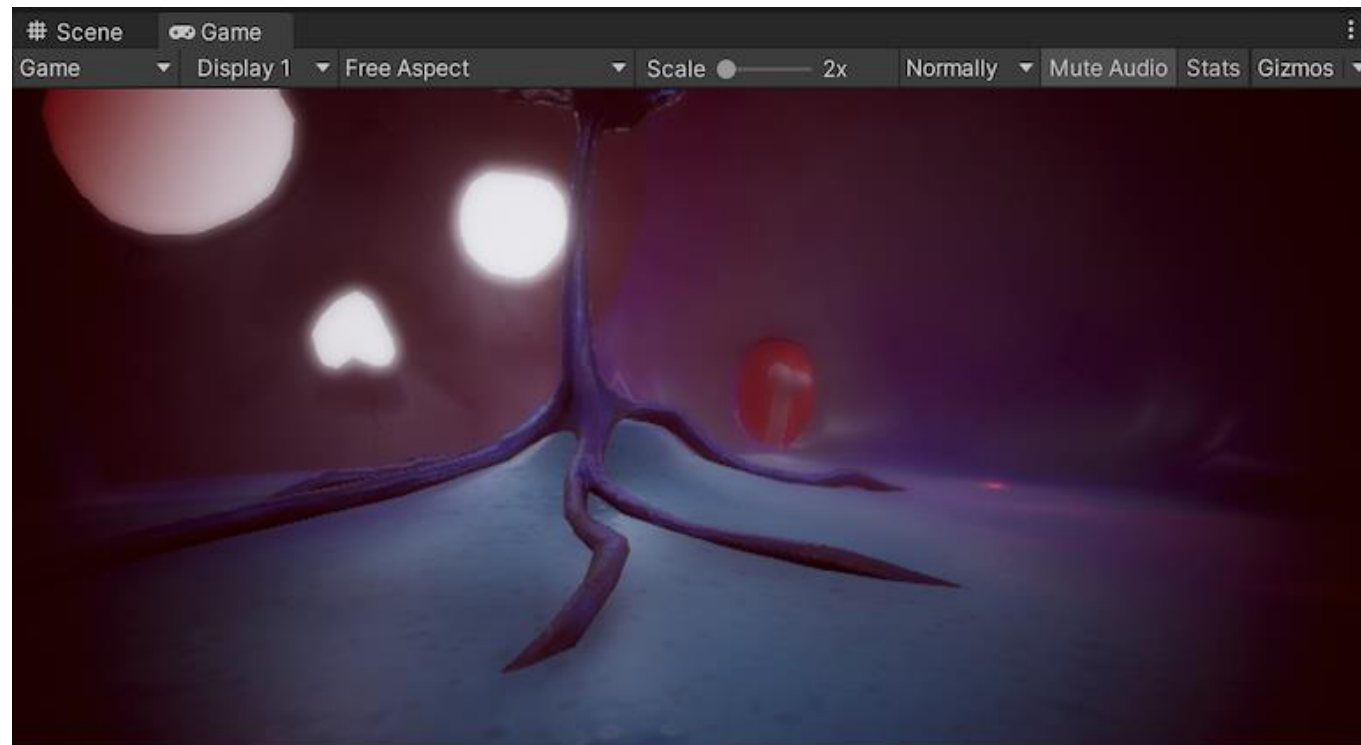
Unity IDE Interface

- **(C) The Game view** simulates what your final rendered game will look like through your Scene Cameras. When you click the Play button, the simulation begins



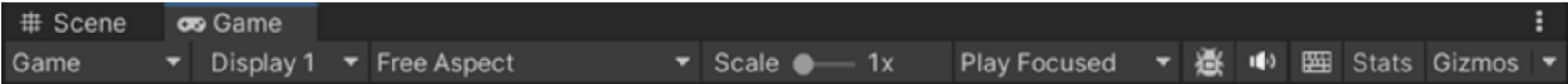
Game View

- The Game view is rendered from the Camera(s) in your application
- It shows how the final, built application looks. You need to use one or more Cameras to control what the player sees when they are using your application



Game View

Game view Control Bar



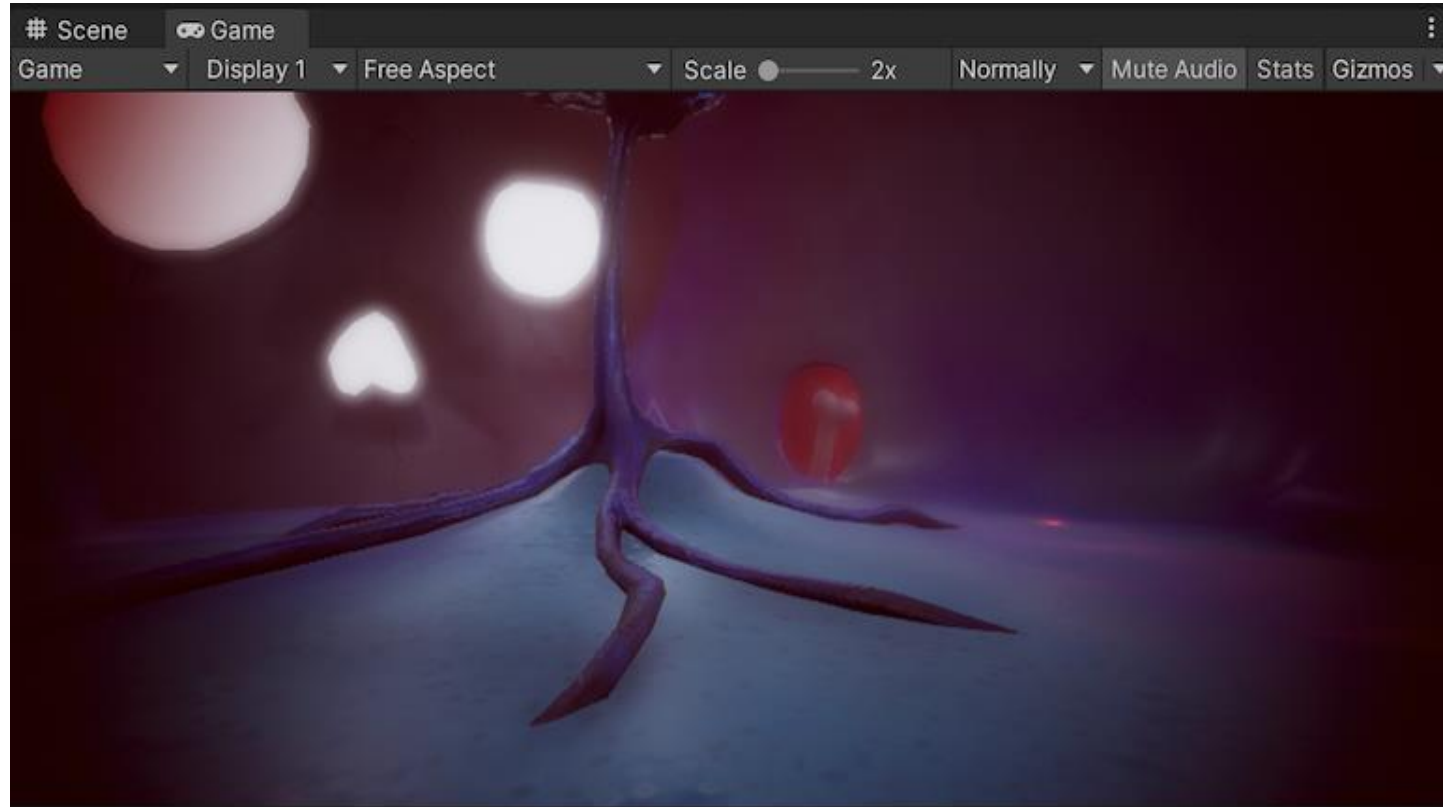
Button		Function
Game/Simulator		Use this menu to enable the Game or Simulator view.
Display		If you have multiple Cameras in the Scene , select the appropriate option to switch between them. By default, this is set to Display 1 . You can assign Displays to Cameras in the Camera module, under Target Display .
Aspect ratio		Select from the available values to test how your game looks on monitors with different aspect ratios. By default, this is set to Free Aspect .
	Low Resolution Aspect Ratios	Enable Low Resolution Aspect Ratios to emulate the pixel density of older displays, which reduces the resolution of the Game view when an aspect ratio is selected. It is always enabled when the Game view is on a non-Retina display.
Scale slider		Scroll right to zoom in and examine areas of the Game screen in more detail. This slider lets you zoom out to see the entire screen where the device resolution is higher than the Game view window size. You can also use the scroll wheel and middle mouse button to do this while the application is stopped or paused.

Game View

Play Mode behavior		This section describes the play mode behavior based on your selection below.
	Focused	Enable Focused to shift focus on the selected Game view while the Editor is in Play mode. Only one Game view can be in focus when you enter the Play mode. Using Maximized mode implies focus on the Maximized Game view. Enabling Focused on a Game view disables it on other Game views.
	VSync (Game view only)	Enable VSync (Game view only) to allow syncing, which is useful when recording a video, for example. Unity attempts to render the Game view at the monitor refresh rate, though this is not guaranteed. When this option is enabled, it is still useful to maximize the Game view in Play mode to hide other views and reduce the number of views that Unity renders.
Enter Play Mode:		Choose from the options below to determine the settings for the Editor when it enters the Play mode.
	Normally	Select this to view the Game view without forcing focus or maximizing any views to full screen.
	Maximized	When this option is selected, Unity runs the Play mode with the Game view maximized to 100% of the Editor window.
Mute audio		Click to enable: use this to mute any in-application audio when you enter the Play mode.
Stats		Click this to show or hide the Statistics overlay, which contains Rendering Statistics about your application's audio and graphics. The overlay lets you monitor the performance of your application while in Play mode.
Gizmos		Click this to show or hide the visibility of Gizmos . To only show certain types of Gizmos in the Play mode, from the Gizmos menu, only enable the Gizmo types you want to see.

Game View

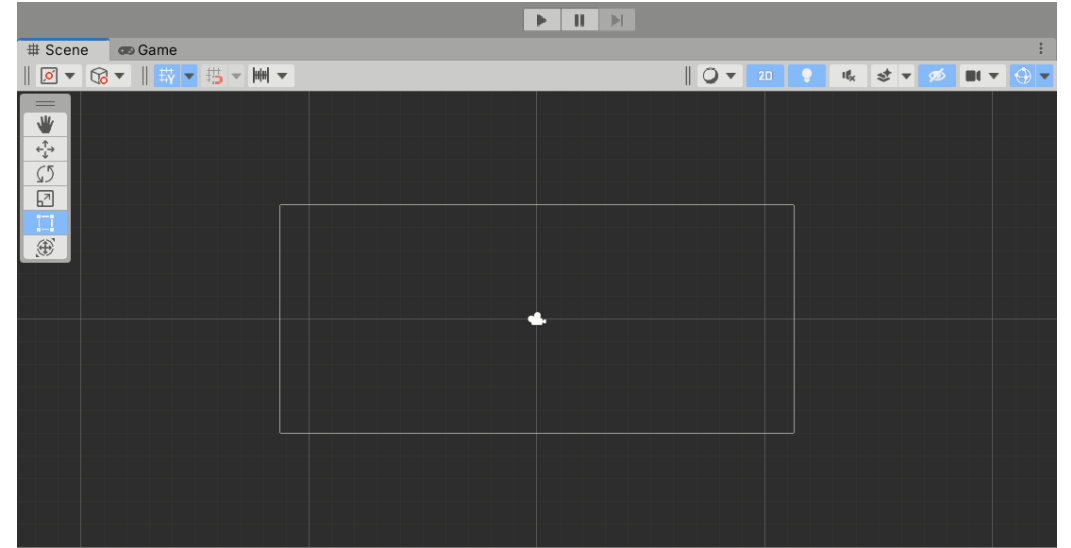
The Game View - Unity Official Tutorials



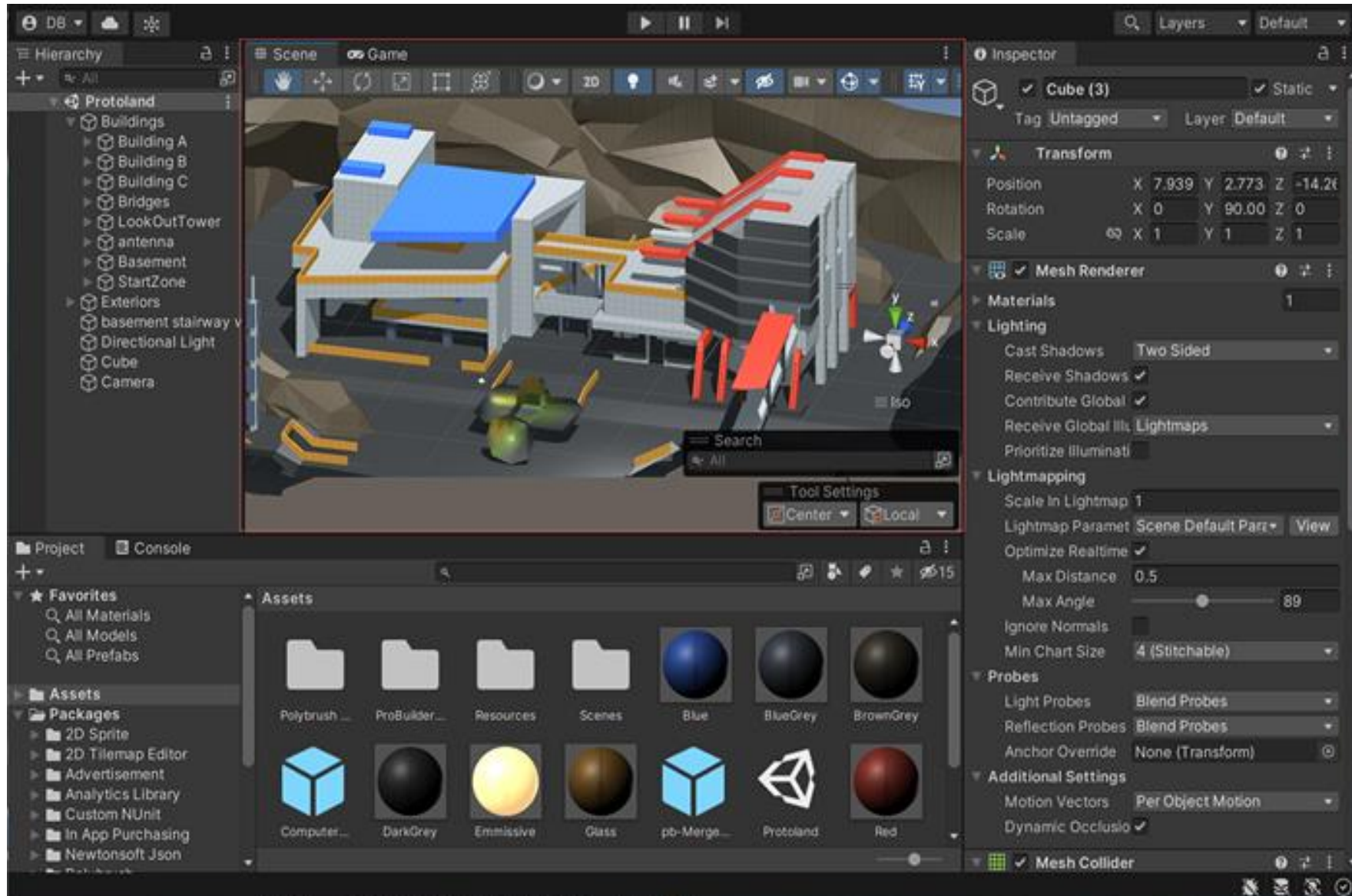
<https://www.youtube.com/watch?v=w7RLUM9TBXY>

Unity IDE Interface

- **(D) The Scene view** allows you to visually navigate and edit your Scene. The Scene view can display a 3D or 2D perspective, depending on the type of Project you are working on



Scene View



Scene View

- The Scene view is where you visualize and interact with the world you create in the Editor
- In the Scene view, you can select, manipulate, and modify GameObjects that act as scenery, characters, cameras, lights, and more

Topic	Description
Overlays	Manage the persistent, customizable panels and toolbars that contain Scene view authoring tools.
Position GameObjects	Modify the Transform values of GameObjects.
Scene view navigation	Move around the Scene view efficiently.
Scene view Camera	Configure the Scene Camera.
Pick and select GameObjects	Select GameObjects in the Scene view and manage their pickability.
Scene visibility	Hide and display GameObjects in the Scene view.
Scene view View Options toolbar	Select view options for the Scene View and enable or disable lighting and audio.
Gizmos menu	Hide and display gizmos in the Scene view.
Custom Editor tools	Create Editor tools to use in the Scene view.

Scene View Navigation

Use the View tool

When the View tool is selected (shortcut: **Q**), the following mouse controls are available:

Control	Description
Pan	 Click and drag to pan the Camera around.
Orbit	 Hold Alt (Windows) or Option (macOS), and left-click and drag to orbit the Camera around the current pivot point. This option isn't available in 2D mode, because the view is orthographic.
Zoom	 Hold Alt (Windows) or Option (macOS), and right-click and drag to zoom the Scene view. On macOS, you can also hold Control, and left-click and drag instead.

Hold down **Shift** to increase the rate of movement and zooming.

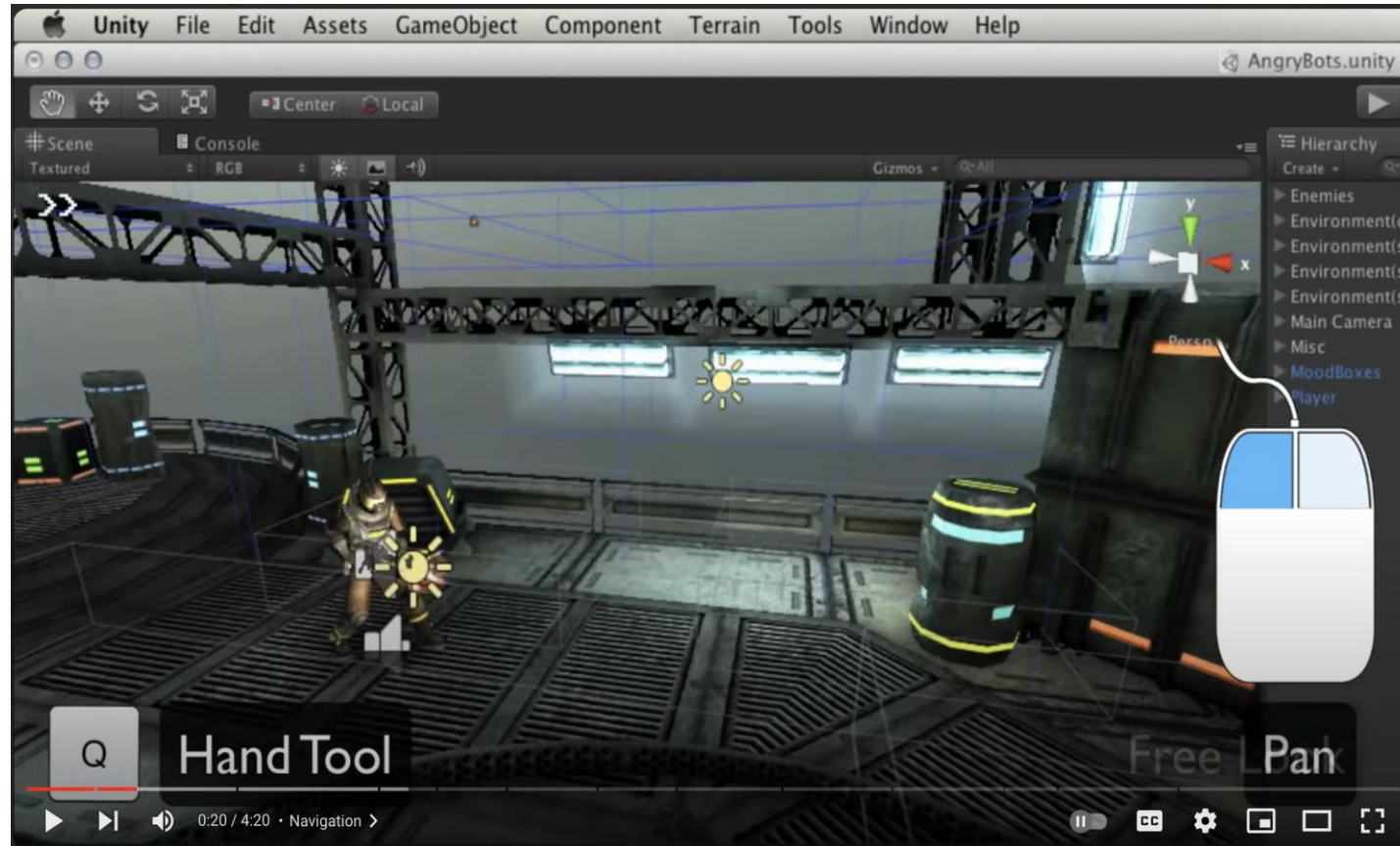
Movement shortcuts

For extra efficiency, these controls can also be used regardless of which transform tool is selected. The most convenient controls depend on which mouse or track-pad you are using.

Action	3-button mouse	2-button mouse or track-pad	Mac with only one mouse button or track-pad
Pan	Hold the middle-mouse button then drag	Hold Alt+Control +left-click, then drag	Hold Option+Command +left-click, then drag
Orbit (Not available in 2D mode)	Hold Alt +left-click, then drag	Hold Alt +left-click, then drag	Hold Option +left-click, then drag
Zoom	Use the scroll wheel, or hold Alt +right-click, then drag	Hold Alt +right-click, then drag	Use the two-finger swipe method to scroll in and out, or hold Option+Control +left-click, then drag
Change speed (only available in Flythrough mode)	Use the scroll wheel while moving.	Drag with two fingers while moving	Drag with two fingers while moving

Scene View Navigation

Scene View navigation tutorial video



[The Scene View - Unity Official Tutorials](https://www.youtube.com/watch?v=fxJR2pHPyTQ)

<https://www.youtube.com/watch?v=fxJR2pHPyTQ>

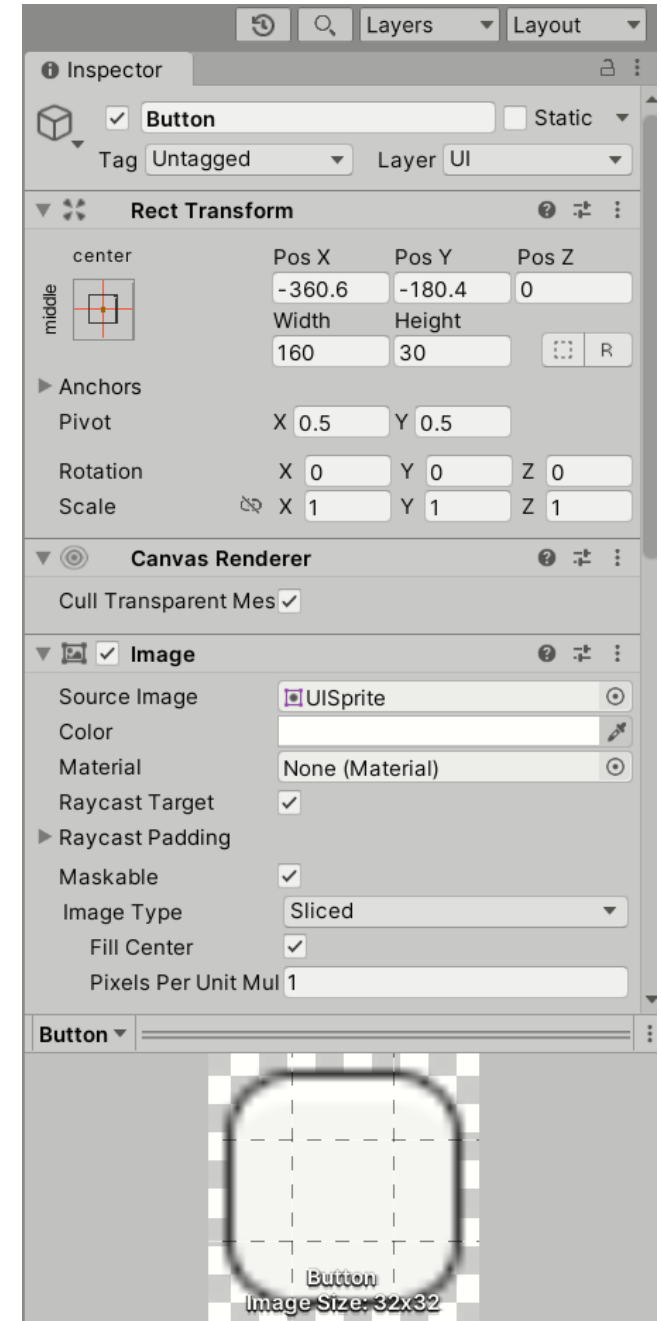
Unity IDE Interface

- **(E) Overlays** contain the basic tools for manipulating the Scene view and the GameObjects within it



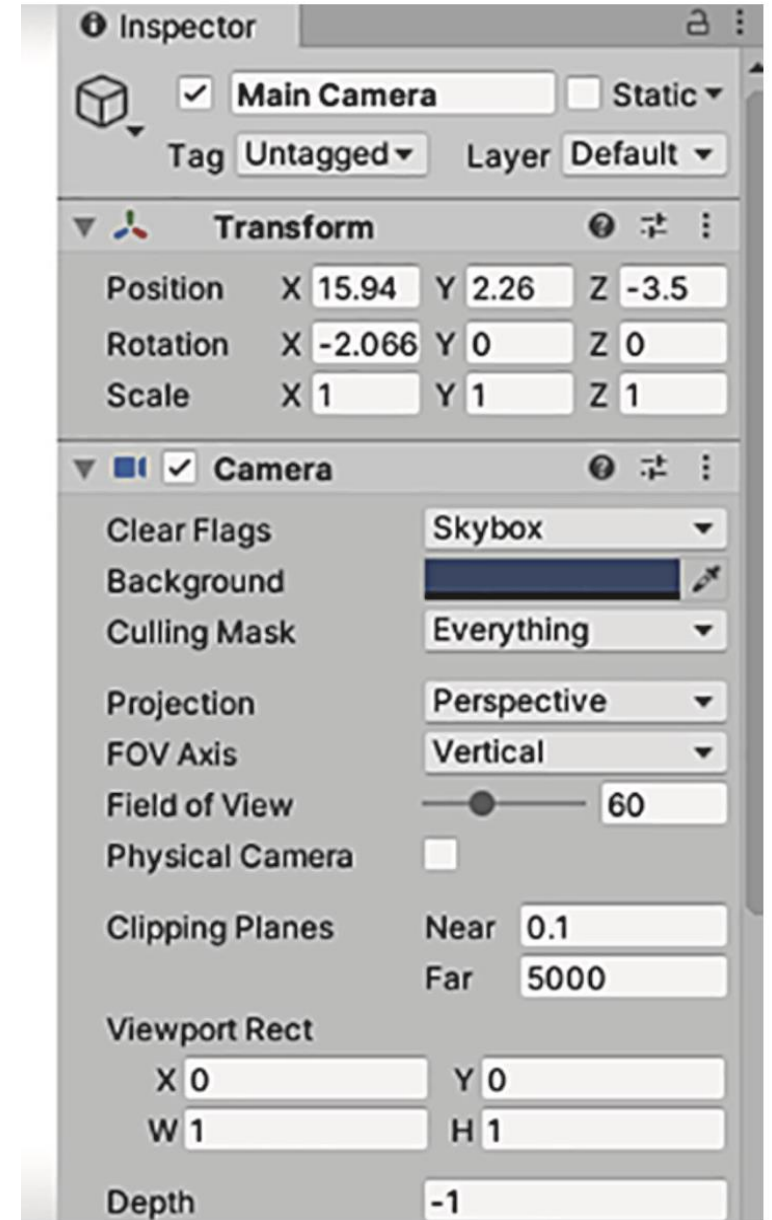
Unity IDE Interface

- (F) The Inspector window allows you to view and edit all the properties of the currently selected GameObject



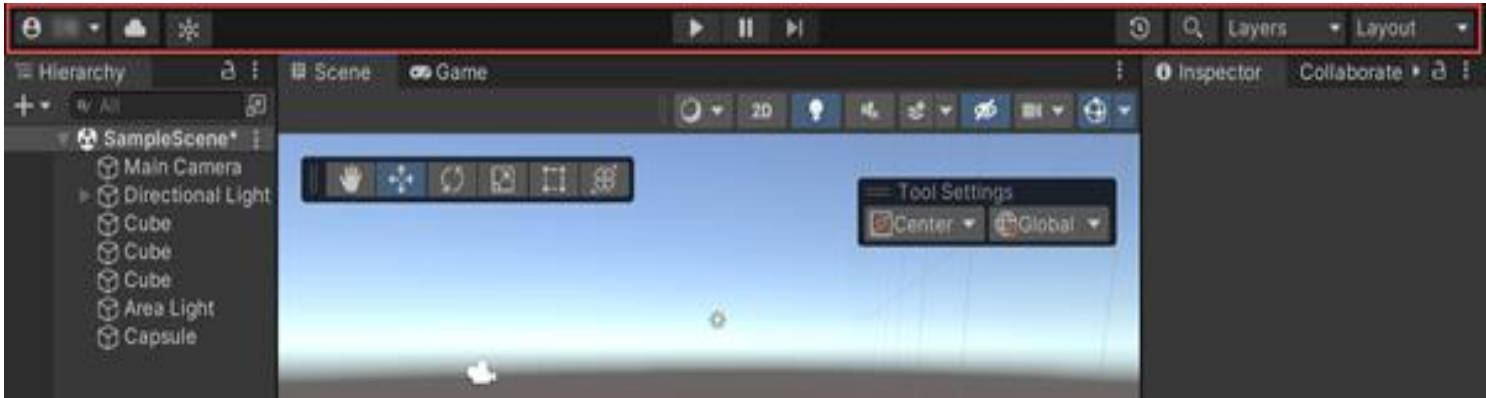
Inspector window

- The **Inspector** shows you information about the currently selected object
- Select an object, and the Inspector is then filled with information about that object. The information shown is pretty much a list of components, and you can even attach or remove components from objects



The Toolbar

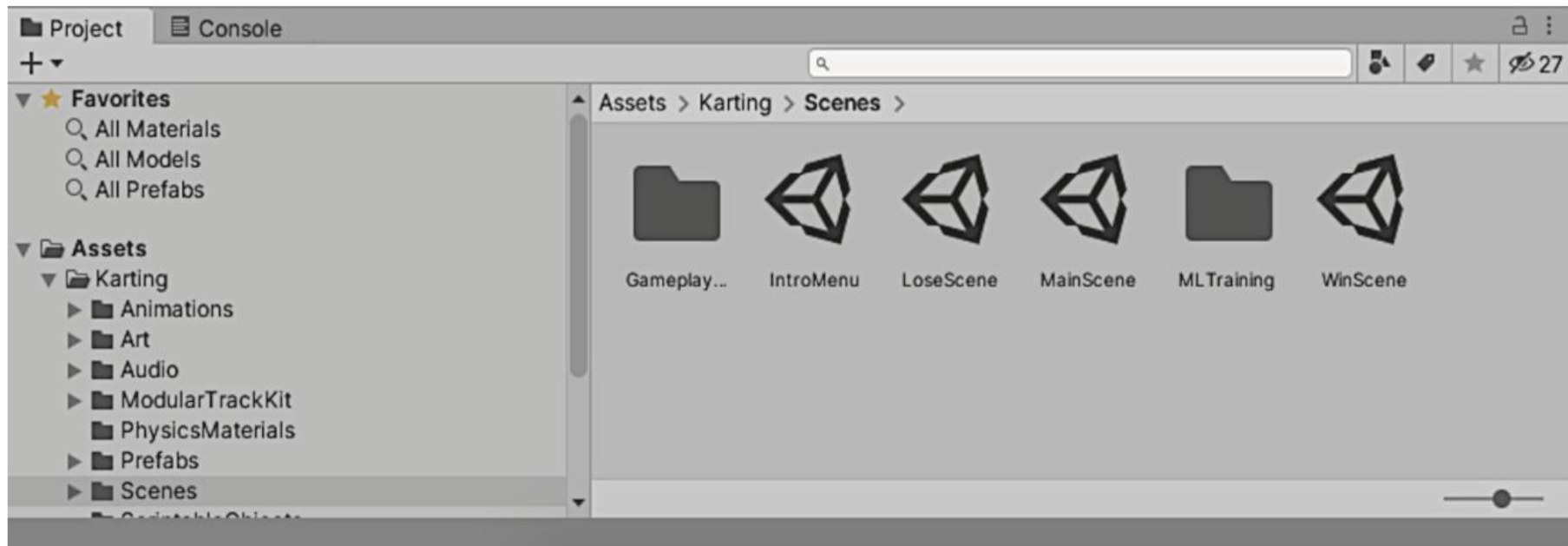
- You can find the Toolbar at the top of the Unity Editor
- It is not a window, and is the only part of the Unity interface that you can't rearrange



Control:	Description:
	Access your Unity Account from the Account drop-down menu.
	Click the Cloud button to open the Unity Services window.
	Use the Play, Pause, and Step buttons in the Game view .
	Use Undo History to view, undo, or redo actions performed in the Editor
	Open Search .
	Control which objects appear in Scene view from the Layers drop-down menu.
	Change the arrangement of your views and then save the new layout or load an existing from the Layout drop-down menu.

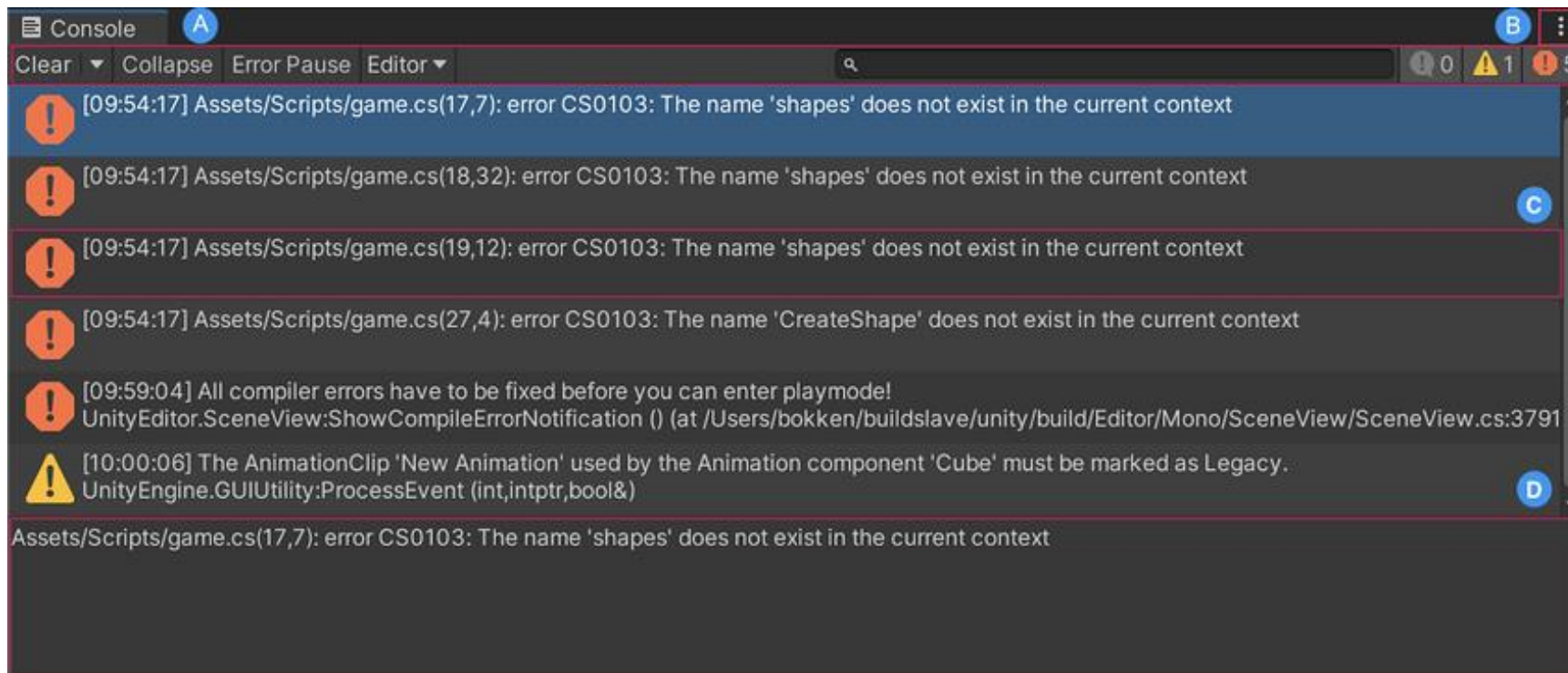
The Project and Console tabs

- Project shows all the assets (art, code, and so on) in the project. Specifically, on the left side of the view is a listing of the project's directories; when you select a directory, the right side of the view shows the individual files in that directory
- Project shows files that may not be contained within any specific scene



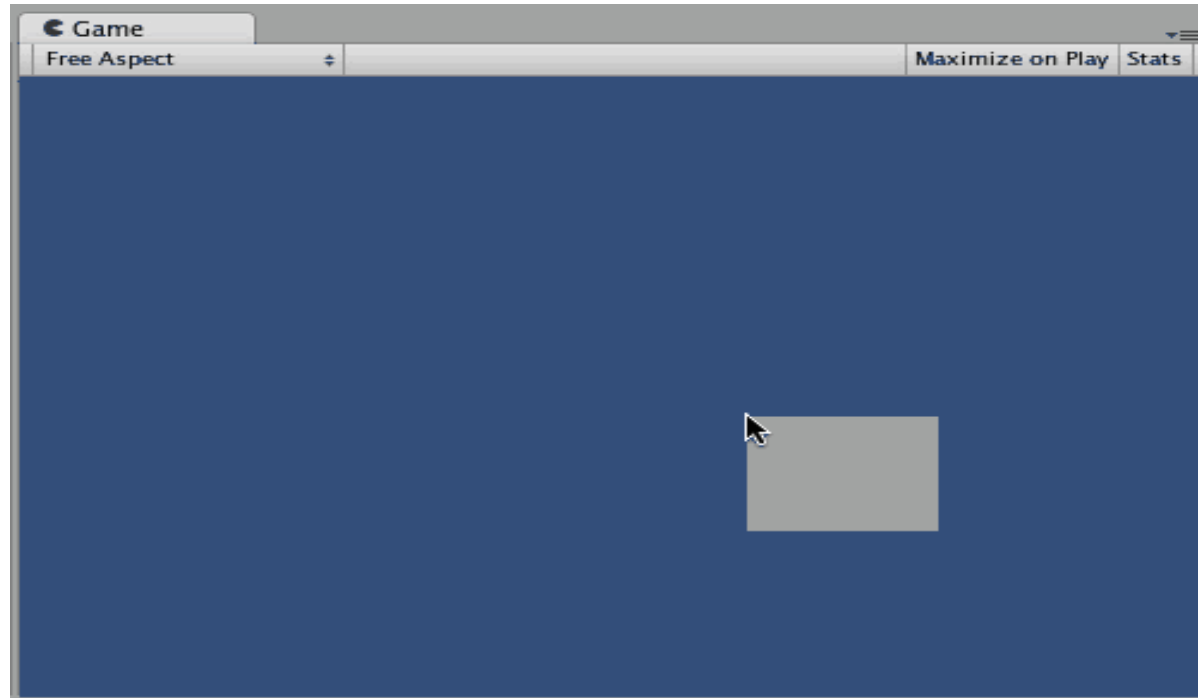
The Project and Console tabs

- The Console tab is the place where messages from the code show up
- Some of these messages will be debugging output that you placed deliberately, but Unity also emits error messages if it encounters problems in the script you wrote



Customizing your workspace

- You can customize your Layout of Views by click-dragging the Tab of any View to one of several locations
- Dropping a Tab in the Tab Area of an existing window will add the Tab beside any existing Tabs. Alternatively, dropping a Tab in any Dock Zone will add the View in a new window

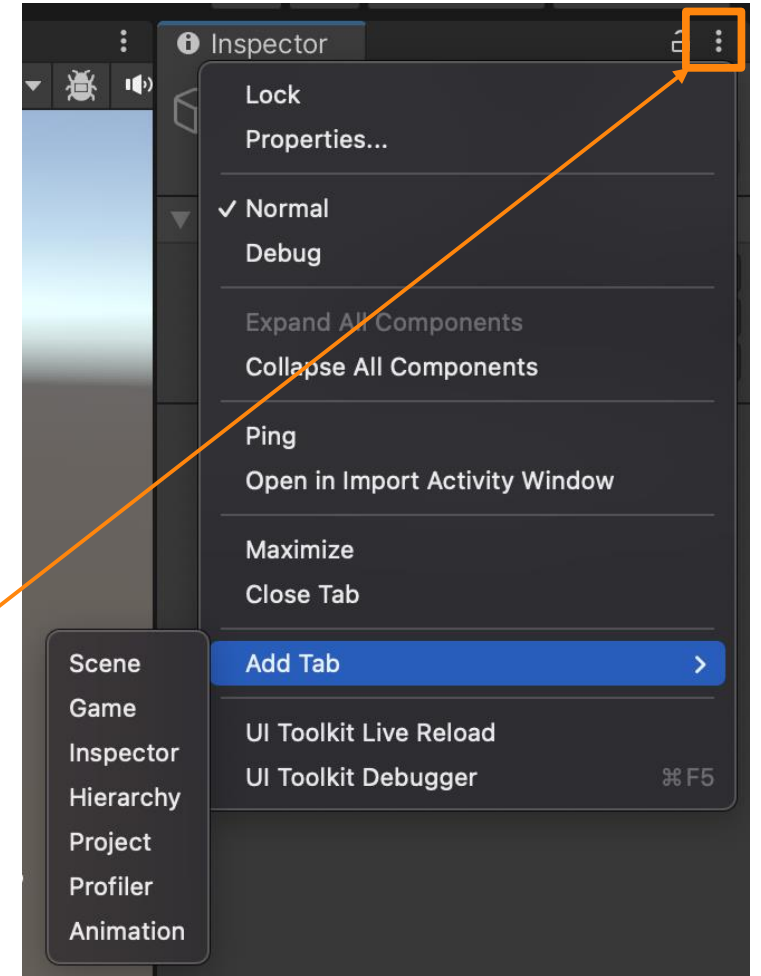


Customizing your workspace



If you can't find a panel on the layout, you can reopen it by going to the **Window > General**, and then choosing the panel you want to add

Or you can click on the **Layout Options** button at the top right of each tab and choose **Add Tab**, to add a new panel



Unity's Scripting Integration

- Unity is deeply integrated with various scripting languages, with C# being the primary language used for scripting game logic within the Unity ecosystem
 - Scripts in Unity play a pivotal role in adding interactivity to game objects.
 - These scripts define the behavior, interactions, and logic behind various game elements, making them dynamic and responsive to player input or events

Unity's Scripting Integration

- Scripts act as the backbone for game interactivity, enabling developers to define how game elements behave and respond in different situations or conditions



The screenshot displays the Unity development environment. On the left, the Hierarchy panel shows a list of scripts attached to a GameObject, with 'Player.cs' selected. On the right, the Console panel shows the C# code for the 'Player.cs' script. The code includes a 'vignette' property, a 'NearHeat' method, and a 'NearSmoke' method, all of which are part of the 'OnTriggerStay' and 'OnCollisionEnter' methods.

```
50 vignette.blur = (1-health) * 2 + smokeEffect * 10 + (health < 0.5 ? 10 : 0);
51 vignette.blurDistance = (1-health) * 2 + smokeEffect * 10;
52 vignette.chromaticAberration = heatEffect * 10;
53 }
54
55
56 void OnTriggerStay(Collider c)
57 {
58     var fire = c.GetComponent<Fire>();
59     if (fire && fire.alive)
60     {
61         float dist = 1-(((transform.position - fire.transform.position).sqrMagnitude));
62         NearHeat(dist);
63     }
64
65     var smoke = c.GetComponent<SmokeParticle>();
66     if (smoke && smoke.GetComponent<Particle>().IsTrigger)
67     {
68         float dist = 1-(((transform.position - smoke.transform.position).sqrMagnitude));
69         NearSmoke(dist);
70     }
71 }
72
73
74 void OnCollisionEnter(Collision c)
75 {
76     var healthBox = c.gameObject.GetComponent<HealthBox>();
77     if (healthBox)
78     {
```

Setting Up Development Environment

- Unity supports the following IDEs
 - **Visual Studio:** When you install Unity on Windows and macOS, by default Unity also installs Visual Studio or Visual Studio for Mac, respectively. By default, the External Script Editor (menu: **Unity > Preferences > External Tools > External Script Editor**) is set to Visual Studio
 - **Visual Studio Code:** Unity supports opening scripts in Visual Studio Code (VS Code). To open scripts in VS Code, go to **Unity > Preferences > External Tools > External Script Editor** and select **Visual Studio Code**
 - **JetBrains Rider:** Unity supports opening scripts in JetBrains Rider. To open scripts in Rider, go to **Unity > Preferences > External Tools > External Script Editor** and select **Rider**

Unity Community and Support

- **Unity Community**

Forum: <https://forum.unity.com/>

Discussions: <https://discussions.unity.com/>

Community: <https://unity.com/community>

- **Unity Manual**

<https://docs.unity3d.com/> (*)

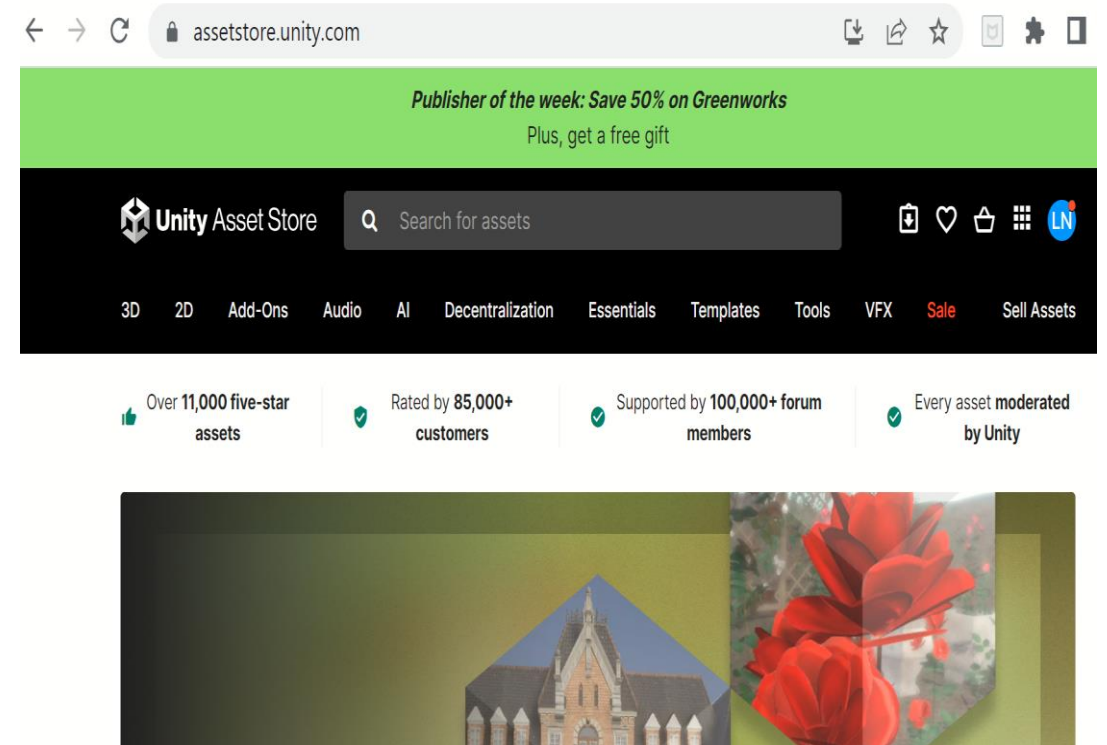
<https://learn.unity.com/>

- **Unity Asset Store**

<https://assetstore.unity.com/>

Asset Store

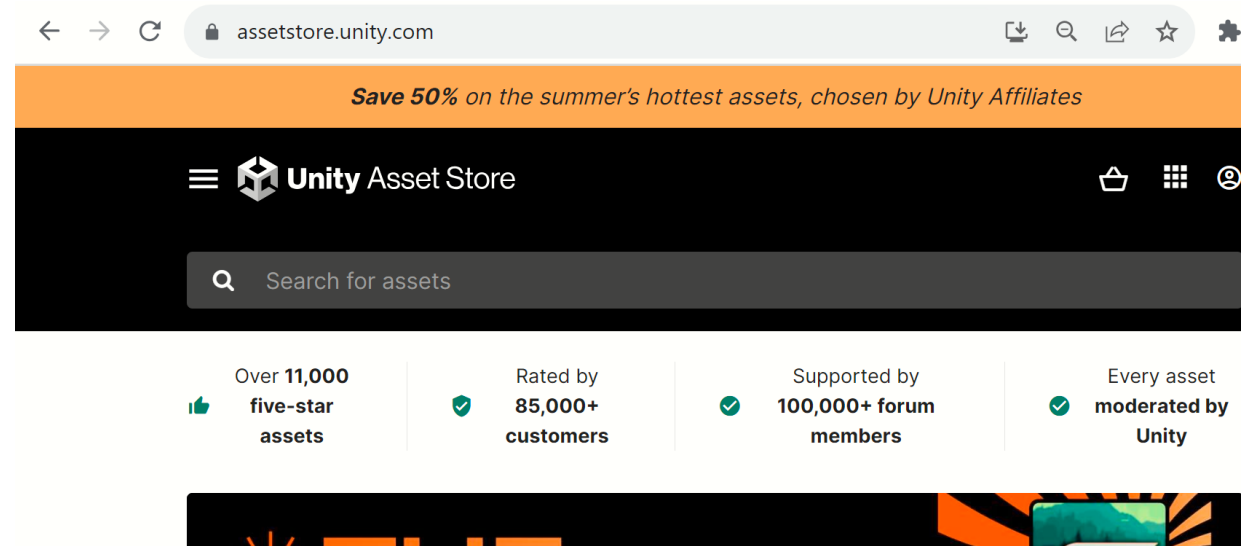
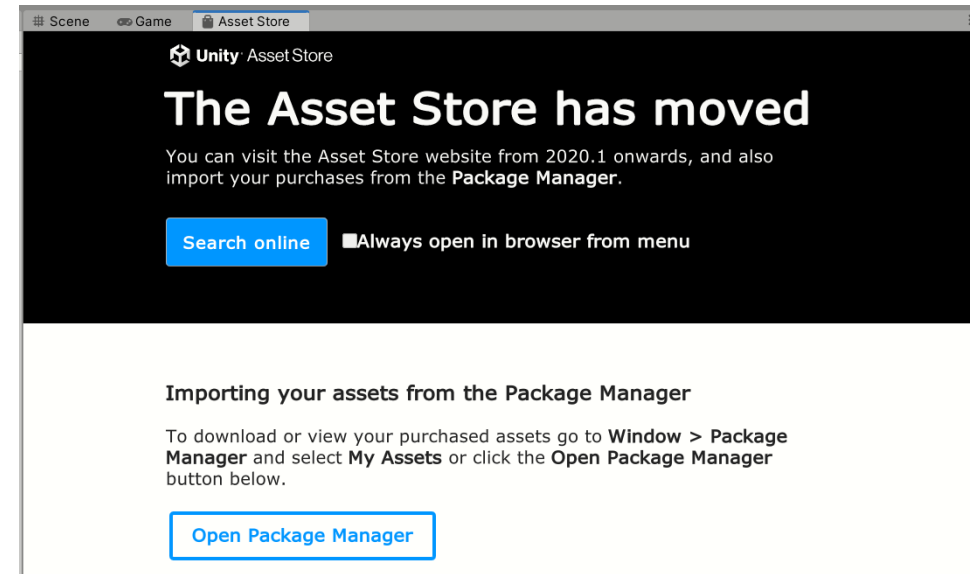
- The Asset Store is an online marketplace where developers can find, buy, and sell digital assets and resources to use in their Unity projects
- It is an integral part of the Unity ecosystem and offers a wide range of content, including 3D models, 2D sprites, textures, animations, audio files, scripts, tools...



<https://assetstore.unity.com/>

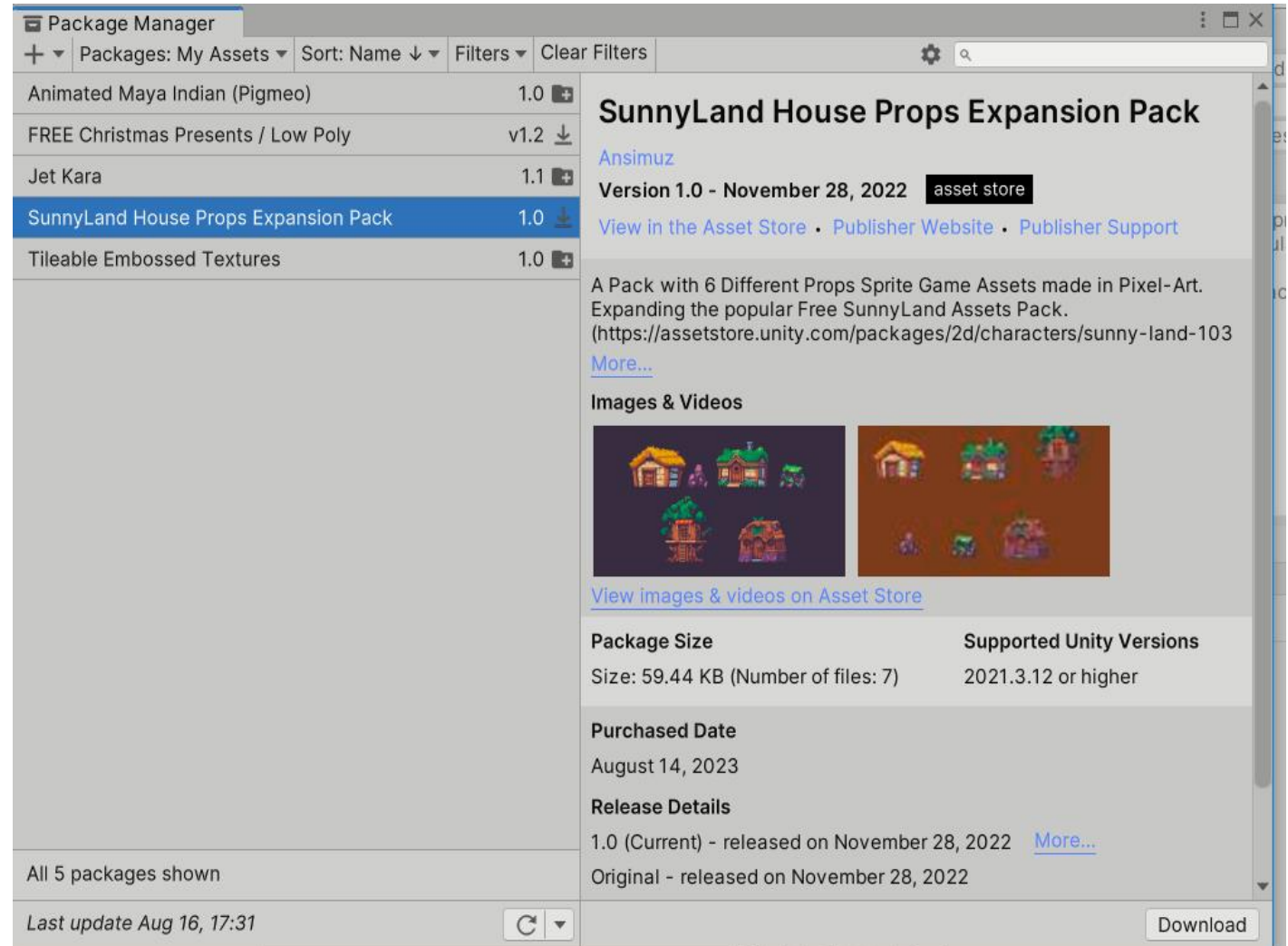
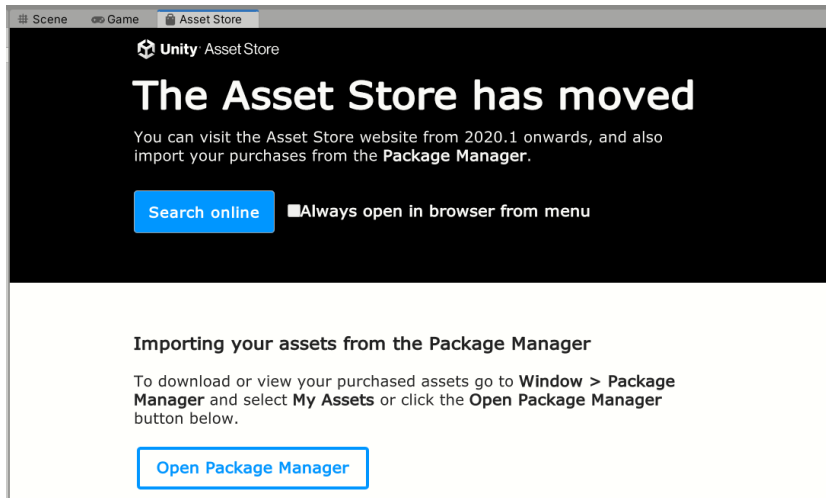
Adding an asset from the Unity Asset Store to project

- In top menu: Select **Window -> Asset Store**
- Click **Search online**, go to website:
<https://assetstore.unity.com/>
- Log in to your Unity account
- Search for assets
- Click **Buy Now / Add to My Assets** button



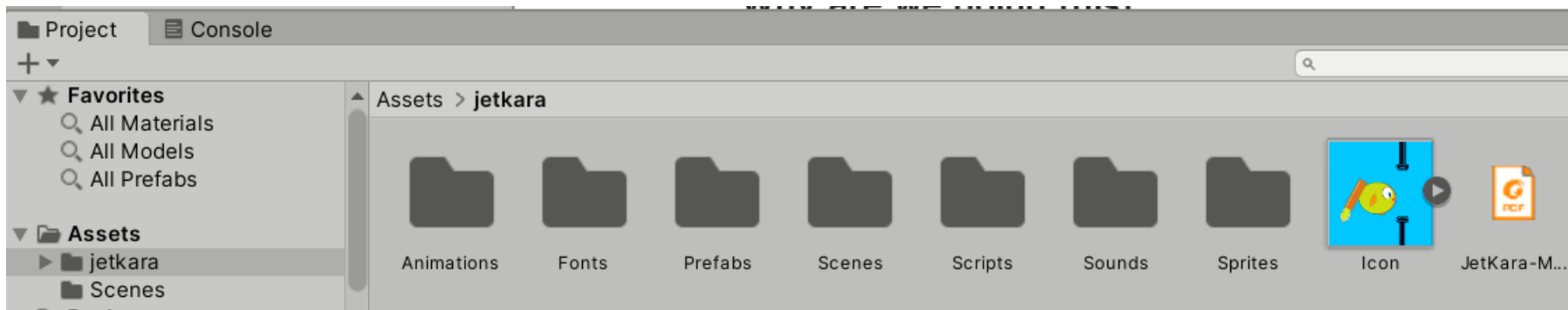
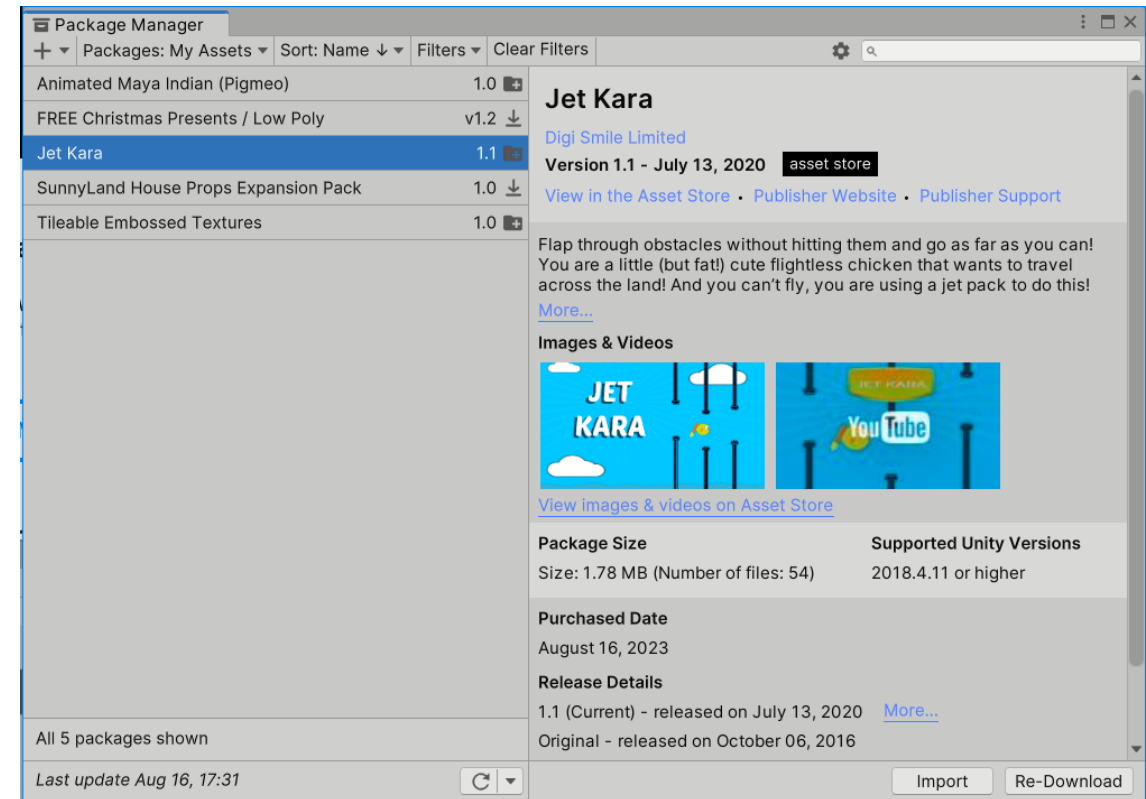
Adding an asset from the Unity Asset Store to Project

- Click **Open Package Manager**
- Select Package from My Assets
- Click **Download**



Adding an asset from the Unity Asset Store to Project

- Click **Import** to import assets from My assets to project



Practice

- **Install tools**

<https://unity.com/download>

- **Review Unity's Interface**

- **Watch video about Scene view:**

<https://www.youtube.com/watch?v=fxJR2pHPyTQ>

- **Watch video about Game view**

<https://www.youtube.com/watch?v=w7RLUM9TBXY>

- **Use Unity Asset store**

<https://assetstore.unity.com/>